Loan Portfolio Risk Analysis: Cohort & Risk Driver Assessment

Understanding performance, risk concentration, and pricing effectiveness

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Executive Summary

- **Project Description** : This analysis evaluates the performance of a U.S. consumer lending portfolio, focusing on loan quality, risk distribution, and portfolio stability.
- **Objective**: Assess overall portfolio health, Identify drivers of loan performance, Evaluate cohort trends and anomaly segments, and Provide actionable recommendations to improve risk and profitability
- **Tools**: Python (Pandas, EDA, Data visualization)
- **Key Metrics**: Total Customers, OS, ENR, TKB30, DPD30 Trend.
- **Analysis Results**:
  - **Cohort Improvement** : Loan performance improved 96.2% (2012) → 98.8% (2019)
  - **Risk Concentration**: Low income borrowers show ~60% higher delinquency and Risk varies by purpose.
  - **Pricing vs Risk:** Mispricing exists
  - **Verification** : Verified borrowers have higher DPD (~12.85 vs 5.49) → indication of selection bias
  - **Geographic Risk**: High exposure 4 top state, South region has the largest portfolio and West region is lowest performance (TKB30 ~97.97%)
- Recommendations:
  - **High Priority**: Tighten approval for low-income segments, Fix mispricing, Reduce exposure to high-risk loan purposes, Reassess verification as risk indicator.
  - **Medium Priority**: Adjust risk in long-term loans, Monitor West region performance

$2.80B

OUTSTANDING (OS)

270.3K

TOTAL CUSTOMERS

98.11%

PORTFOLIO TKB30

Business Context

RevoVin is a pioneering financial technology company that specialized in providing innovative lending solution to individuals and small business.

RevoFin operate through seamless online platform that connects borrowers with a range of loan options. By streamlining the application process, utilizing automated credit assessments and offering competitive interest rate, RevoFin positioned themself as a preferred lending partner.

Our revenue is designed around transaction fee, interest income, and strategic partnership with business seeking to provide financial product to their customers.

Objective

1. Assess the stability of the $2.8B outstanding loan book.
2. Identify which customer attributes (Income, Home Ownership, Employment) and loan characteristics (Term, Purpose) drive delinquency (DPD).
3. Determine if newer loans are safer than older ones.
4. Evaluate if granting loans to "Not Verified" customers is a risk or a strategic advantage.

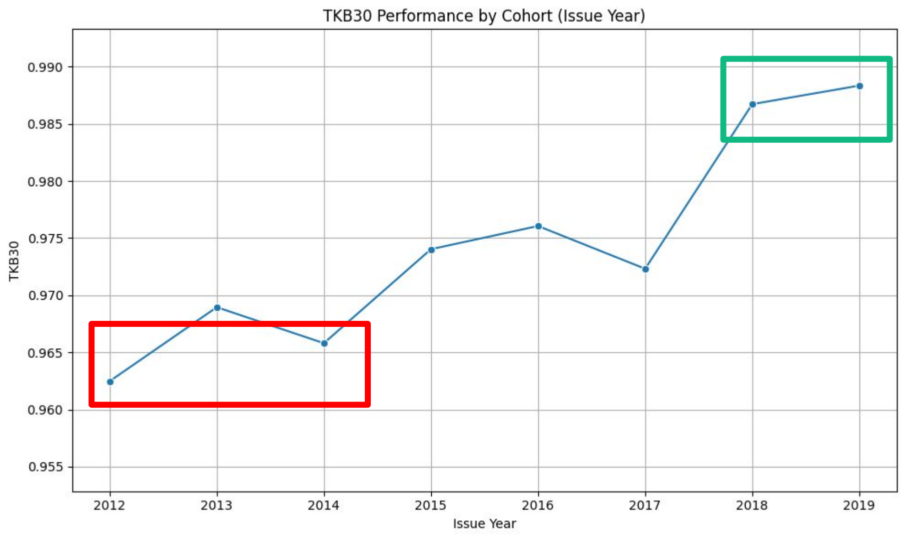
Hypothesis

1. Newer cohorts have better loan performance
2. Risk is concentrated in certain loan purposes and borrower segments
3. Higher interest rates compensate for higher risk
4. Verification status indicates lower risk

Methodology

- Business Understanding
- Cleaning Data (customer_id)
- Data Analysis and EDA (generate insight TKB30, ENR, DPD, Risk, Verification, State, Region)
- Presentation

Strong improvement across cohorts



2012–2014 are "Anomaly Cohorts" with lower TKB30 (~96-97%).

- TKB30 moving from 96.24% (2012) → 96.5% (2014) → 98.83% (2019)
- DPD drops: 3.75% → 1.16%

Improvement = +2.6% TKB30 → Big in lending

2012 cohort was an "anomaly" because we were charging high interest rates (18.3%) to lower-income groups ($76K).

Anomaly Cohort Profile

	2012	2014	2019
Average Income	$76.1K	$74.1K	$84.8K
Interest rate	18.3%	13.7%	12.8%
Ownership	63.4%	60.9%	60.8%

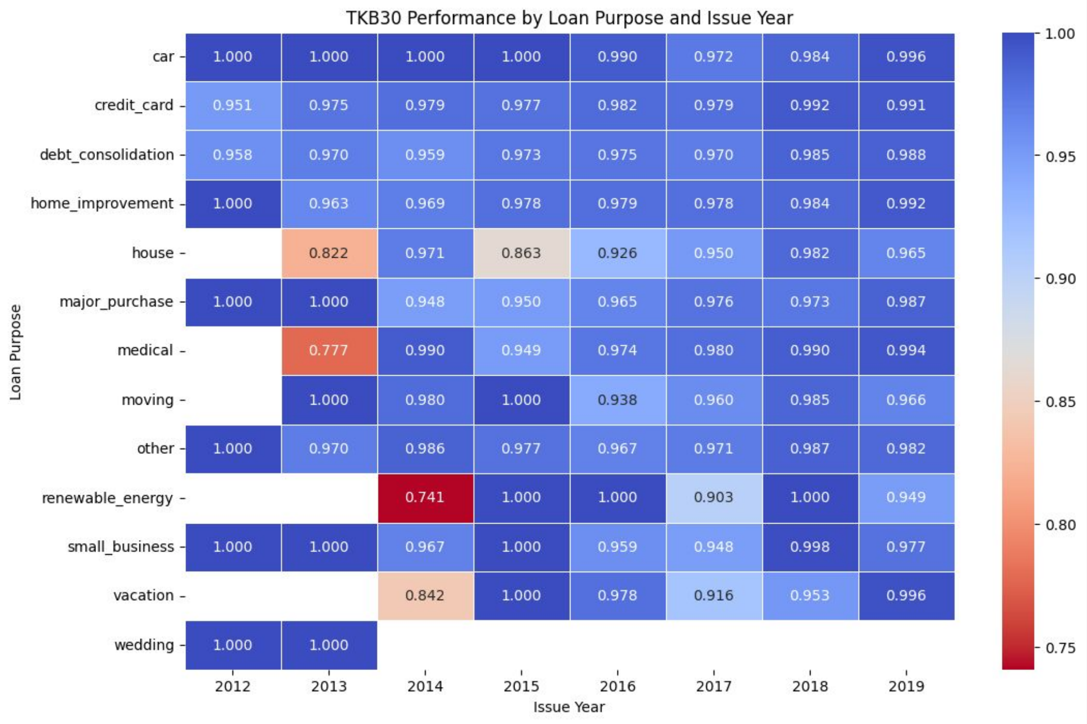
Interest rate difference: ~5.5%

- Rate ↓ 5.5%
- Income ↑ ~10%

Recommendation :

- Focus on improving borrower quality + pricing optimization to drive better cohort performance
- Tighten approval for low-income segments

Risk is not evenly distributed.



High risk:

- Renewable energy (95.7%)
- Vacation (96.0%)

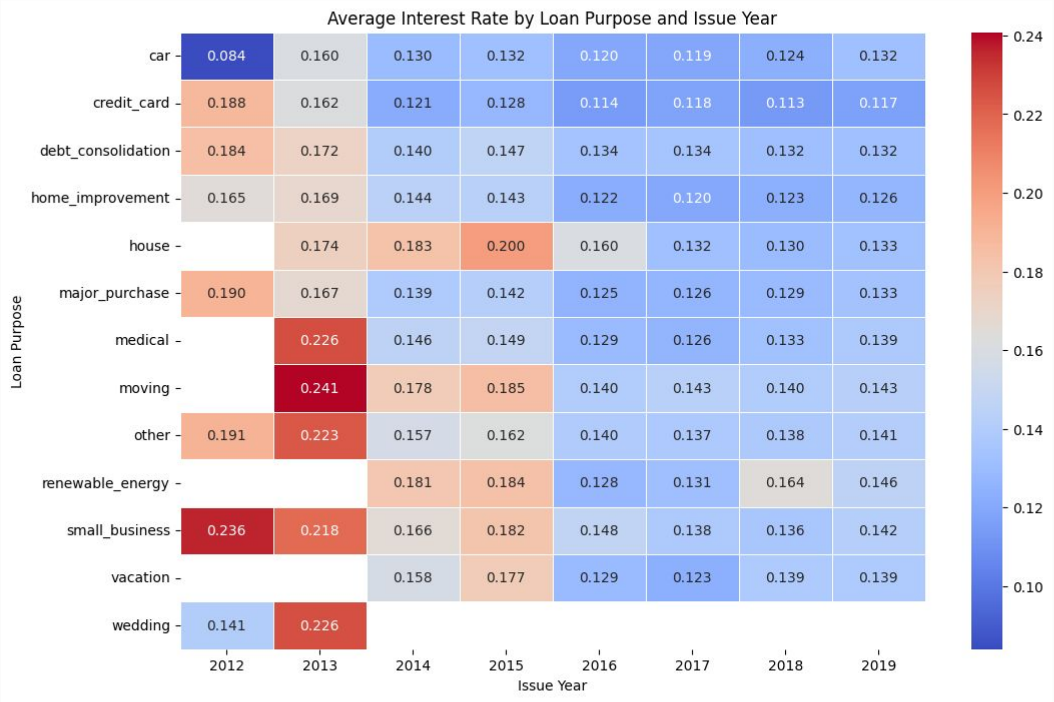
Low risk:

- Credit card (98.6%)

Gap: ~3% difference — Huge in lending

Certain loan purposes consistently show weaker performance, indicating behavioral risk differences among borrowers

Higher-risk segments have higher rates

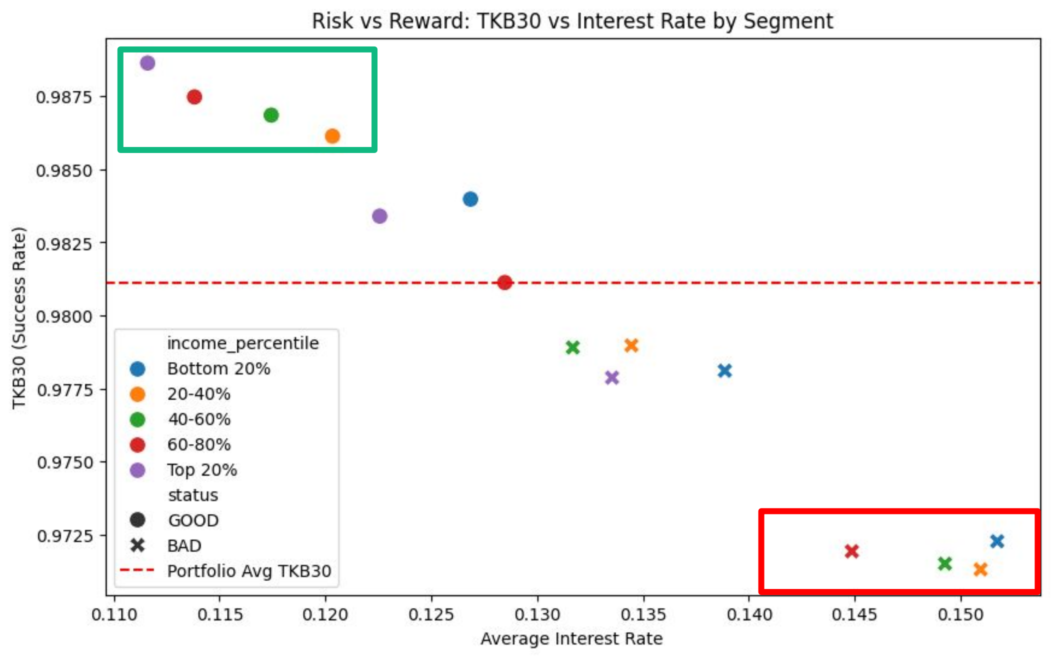


- Renewable energy: 14.7%
- Credit card: 11.6%
- Spread: ~3%

Categories like small business, moving, and medical tend to carry higher rates compared to safer categories like credit cards.

This suggests that our pricing model is aligned with risk—at least at a high level

Some high-rate loans still underperform



- High rate (~15%) but TKB30 ~97.1 (bad)
- Low rate (~11%) but TKB30 ~98.8 (good)

Ideally, higher interest rates should compensate for higher risk. But in this chart, there's segments with both high interest rates and lower TKB30 performance.

This indicates that in some segments, we are taking on higher risk without sufficient return.

Low Income vs High Income

	DPD	Average Rate	TKB 30
Low Income (Percentile 20%)	10.4	13.9%	97.8%
High Income (Top 20%)	6.5	12.1%	98.4%

- TKB30: 97.8% vs 98.4% → worse repayment
- DPD: 10.4 vs 6.5 (~60% higher)
- Rate gap = only ~1.8%

Recommendation:
- Tighten approval low income
- Adjust pricing by income
- Add income threshold filter

Verification Analysis

verification_status	dpd	count
Not Verified	5.49%	97.8K
Source Verified	9.09%	105.3K
Verified	12.85%	67.1K

- Not Verified TKB30: ~98.8%
- Verified TKB30: ~97.1%
- DPD difference: 2x higher

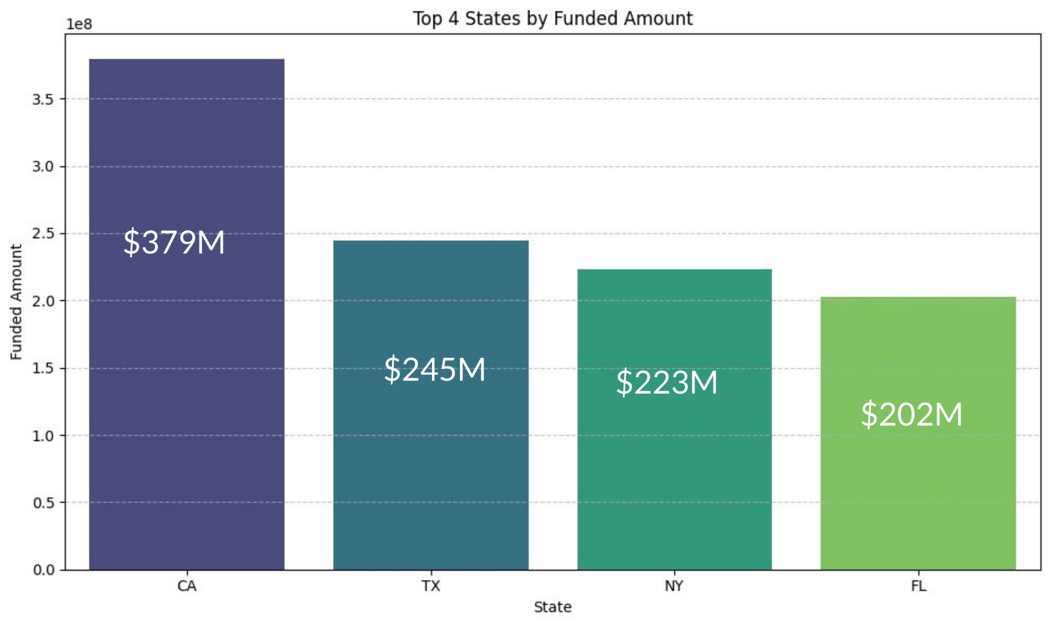
Critical Attention Needed!

We are potentially over-verifying low-risk customers and under-scrutinizing high-risk ones.

Recommendation:

- Re-evaluate verification logic
- Don't use as primary risk signal

Portfolio Concentration by 4 Major State



A significant portion of our portfolio is concentrated in a few key states, particularly California, Texas, New York, and Florida.

This creates concentration risk, where performance deterioration in these regions could disproportionately impact the overall portfolio

Risk Distribution by Region

Region	OS	TKB30	Share
South	1.02B	98.09%	36.6%
West	701M	97.97% (lowest)	25.0%
Northeast	573M	98.13%	20.5%
Midwest	498M	98.28%	17.8%

- West TKB30 = 97.97% (lowest)
- South share = 36.6% (highest)

This indicates that both high exposure and lower-performing regions require close monitoring

Recommendation:

Monitor top states closely
Improve risk control in West
Diversify exposure
Add regional risk monitoring

Key Takeaways

Portfolio performance is strong, but risk is driven by:

- Cohort quality improvements
- Borrower income and loan purpose
- Incomplete pricing optimization
- Misleading verification signals
- Geographic concentration

Recommendation

High Priority :

- Fix pricing gaps
- Tighten low-income loan
- Optimize by reducing risky purpose
- Improve process verification

Medium Priority :

- Adjust risk long-term loans
- Manage regional exposure

Appendix

Calculation : Google Collab

Term 36 Months VS 60 Months

	avg_int_rate	total_os	dpd30_os	TKB30
term				
36 months	0.118102	1564925050	26733500	0.982917
60 months	0.152210	1236153800	26308750	0.978717

The 60-Month Trap: 60-month loans have a higher average interest rate (15.2% vs. 11.8%). In Fintech, we charge more for longer loans because there is a higher chance something will go wrong over 5 years compared to 3 years.

- The Risk Gap: The TKB30 for 60 months (97.87%) is lower than 36 months (98.29%).
- Meaning: Even though you are charging more interest (making more money per loan), these customers are failing to pay more often.
- The "Why": A 0.42% difference might look small, but on a 1.2 Billion portfolio, that is millions in potential losses.

Risk by Home Ownership

	total_os	dpd30_os	TKB30
home_ownership			
RENT	962962300	22013150	0.977140
OWN	308016275	6272950	0.979634
ANY	4381575	74000	0.983111
MORTGAGE	1525718700	24682150	0.983823

- RENT (0.977): Renters have the lowest TKB30. This is common because renters are more mobile and typically have fewer assets. They represent a large chunk of your portfolio (~Rp962M), making them your primary risk area.
- MORTGAGE (0.983): These are your "star" customers. Someone with a mortgage is tied to a property; they have a high "cost of failure" if they stop paying.
- Takeaway: If you want to improve your overall TKB30, you should look closely at how you approve "Renters."